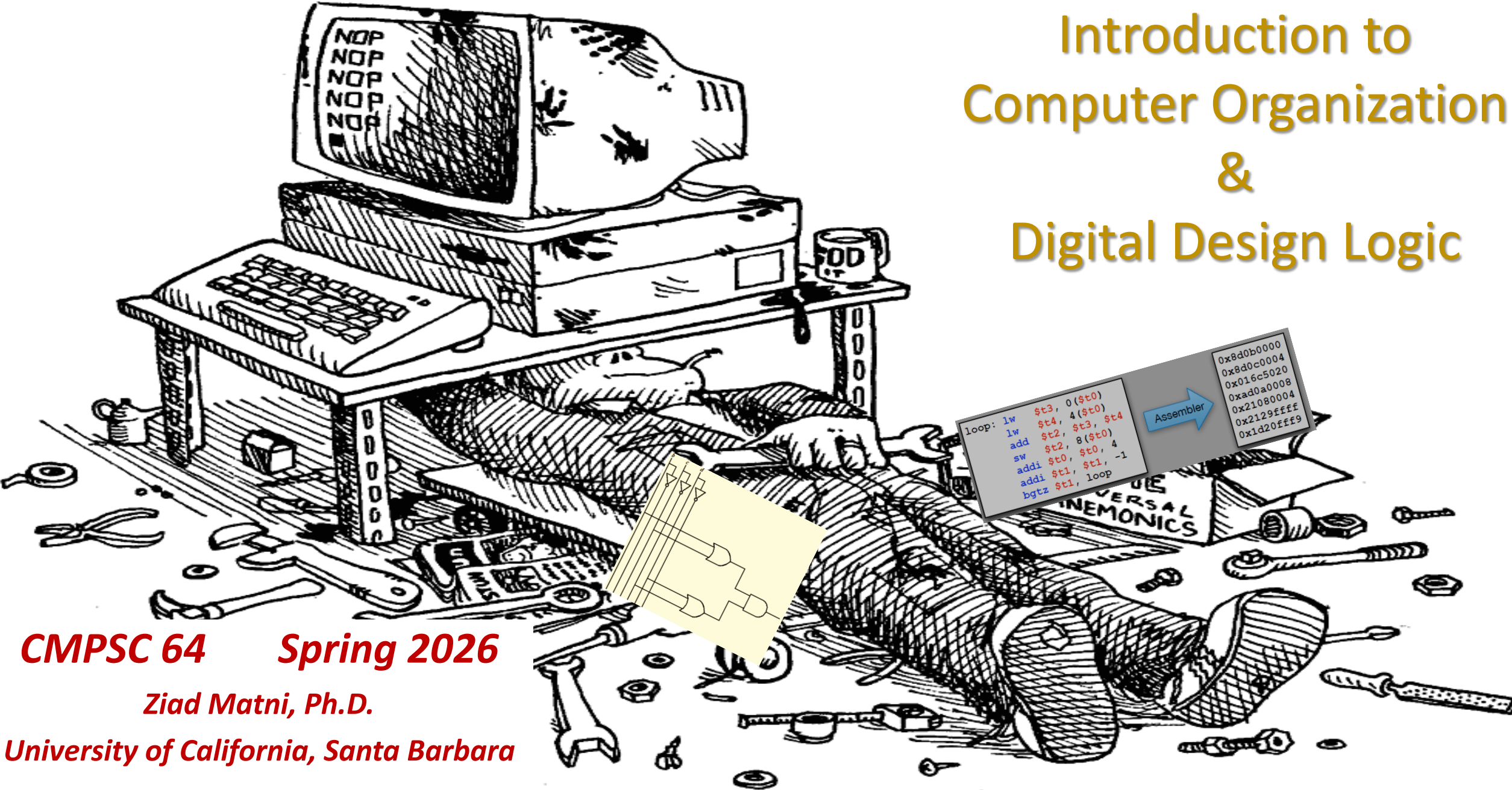


Introduction to Computer Organization & Digital Design Logic



CMPSC 64 Spring 2026

Ziad Matni, Ph.D.

University of California, Santa Barbara

Your Instructor

Ziad Matni, Ph.D.

(zee-ahd mat-knee)

Assistant Teaching Professor in Computer Science



About me...

What to Call Me?? Prof. Matni!

Teacher, Researcher, Engineer, Coffee Connoisseur

The CS 64 (S26) Instructional Team

Teaching Assistants (TAs)

*Run the labs, grading
Hold office hours
Graduate students*



Kali Hale

Rasta Tadayon

Learning Assistants (LAs)

*Help with the labs
Hold office hours
Undergraduate students*



Riya Gupta

Ivy Holiday

CONTACT DETAILS & OFFICE HOURS ARE ON **CANVAS and **SYLLABUS**!**

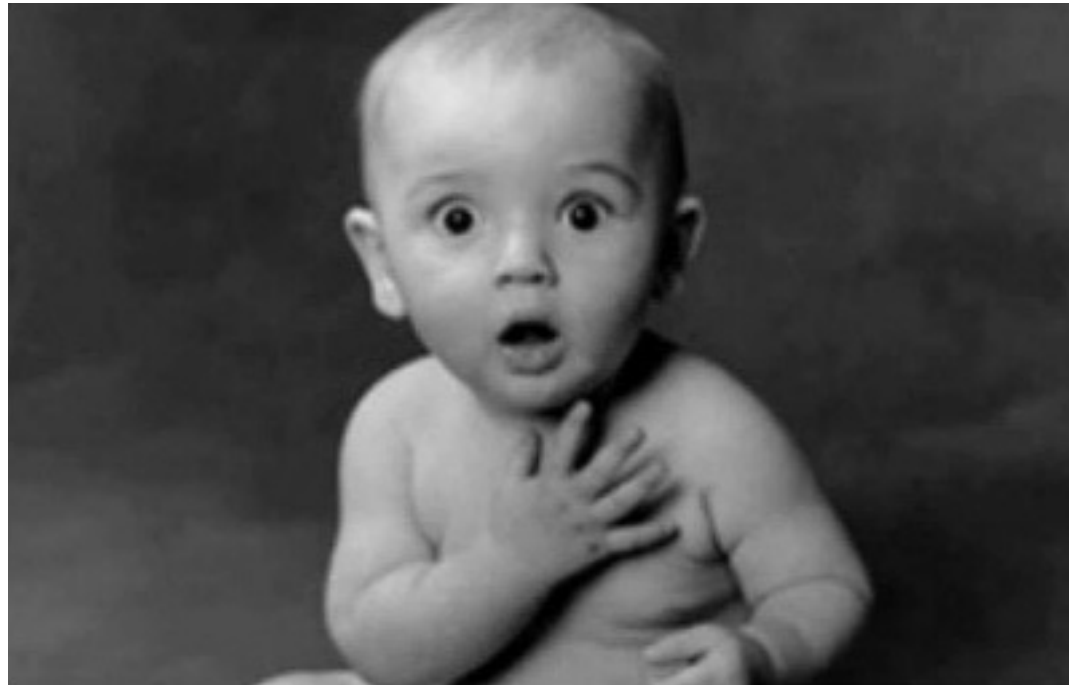
How About You?!

Who are YOU?!?!?!?

- *Year? Major?*
- *Did you enjoy CS16, especially **pointers** & **memory** stuff?*
- *Have you ever taken apart or put together a computer?*

Also:

- 1. Trying to add the class?*
- 2. Trying to switch labs?*
- 3. We're taking attendance...*



The Trifecta of Success in this Class!

Lectures and
Reading Material

Weekly
Assignments

Class
Participation

Your
Lab/Sections

Prof's, TAs'
Office Hours

Online Help
via Piazza

Quizzes & Exams



Where is Everything? What Should I Get?

- You NOW have accounts ready on the following (i.e., I have registered you):

- Canvas
- Gradescope
- Piazza

IMPORTANT: I used your UMAIL addresses, i.e., **gaucho@umail.ucsb.edu**

PLEASE CHECK THAT YOU HAVE ACCESS!!!

- **Software for this course:**

- iClicker
- You need access to your CoE CSIL accounts and the use of UNIX terminal

CS 64 FAQs

- Where is everything (info-wise)?
- Does Prof. Matni take attendance in lectures?
 - What are the lab sections about?
Do I have to go to them?
 - Quizzes??? Exams??!?!?! (gulp!)
- How are the homework assignments setup?
 - What are the grading policies here?
 - Textbook – yay or nay?
 - If I need HELP, what do I DO?!
 - Ok – tell me more about this course...
- Canvas!
- Only in Week 1, and theeeen...
- Concept reviews, homework help, demos, etc.
Yes – we DO take attendance for labs, buuut...
- 5 Quizzes (drop one) + 1 Final Exam
- Homework is submitted on Gradescope...
- See syllabus (also, see next slide)
- Yes (recommended) + prof's readings (important)
- Ask questions! In class, in lab, in o/hs, Piazza...
- Comin' right up!...

Grading Policies (See Details in Syllabus)

- Grading

- Participation: 2%
- Assignments: 28%
- Quizzes: 30%
- Final: 40%

- Late Policy

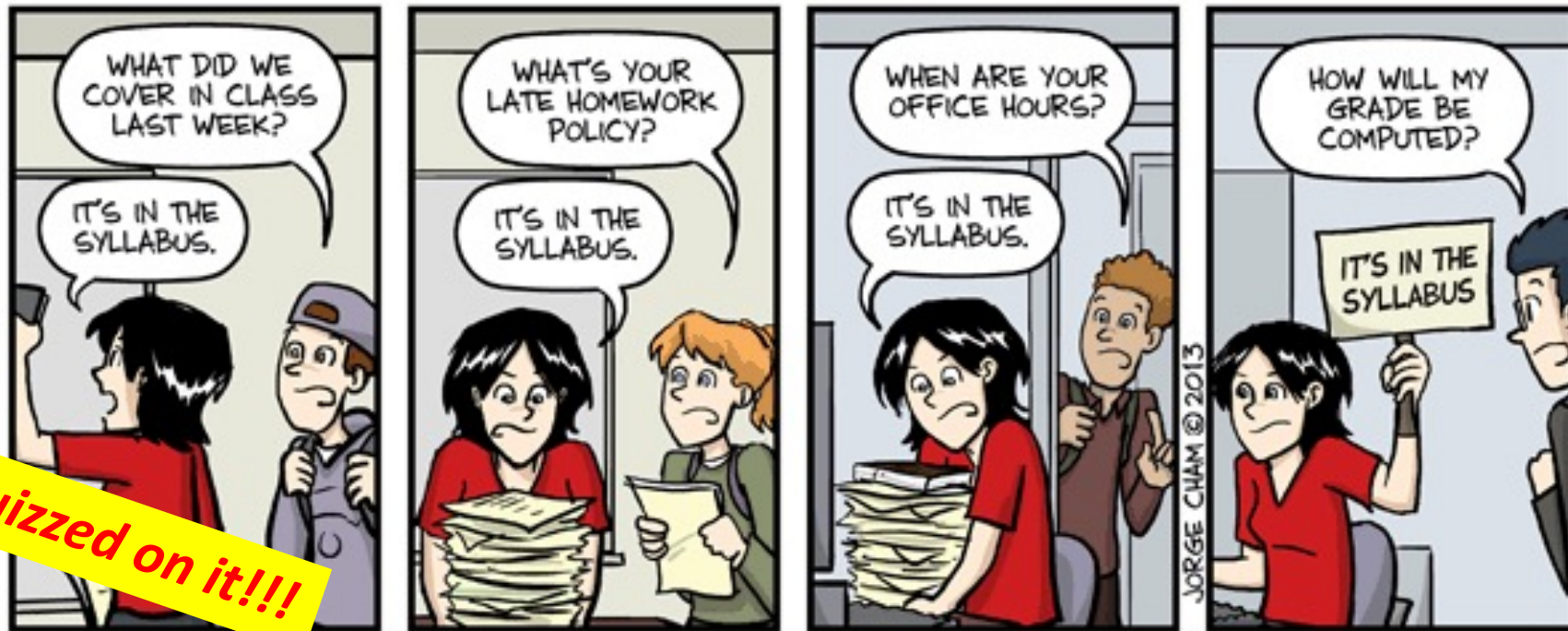
- Assignments turned in after due, within 24 hrs: **Lose 20%**
- No late assignments accepted after 24 hrs: **zero grade**

My Policies (short summary – details in syllabus)

- I expect you to do your OWN work at ALL times
- **I take Cheating and Plagiarism VERY seriously**
 - What happens if you work with someone on an assignment?
 - What happens if you find a solution online somewhere? Or use chatGPT?
 - What happens if you use an AI to help you with your assignment?
 - Bottom line: YOU are responsible for your learning and your ETHICAL behavior
- What about using LLMs (i.e., Chat GPT, Gemini, others...)?
- UCSB's Academic Integrity and Honesty Policy = My Policy



Just in Case...



IT'S IN THE SYLLABUS

This message brought to you by every instructor that ever lived.

WWW.PHDCOMICS.COM

Lab/Discussion Sections

- TA will run the lab sections
 - **Fridays – in person!!**
 - Course material review, work on the assignment
 - Attendance is taken (if you skip more than 2 labs, it'll start affecting your grade)
- We **STRONGLY encourage** you to ask the TA questions to help you bridge any “knowledge” gaps
 - i.e., from what you may have not fully understood in lecture or readings.

Exams

- Quizzes on **Wednesdays of Weeks 2, 4, 6, 8, and 10**** (** Monday)
 - In-Class (at the start, i.e., 11 AM sharp), In-Person, NO MAKE UP
 - *I will drop your lowest quiz grade at the end of the course*
- Cumulative Final Exam on **Thursday, June 11 at 12:00 PM**
 - In-Class, In-Person, NO MAKE UP
- Again: **NO make up for missed quizzes or final exam**

Format of Assignments

- Given every Monday
- Discussed in lab on Friday
- Due by following Monday*

** There might be some exceptions*

- This class will have **weekly assignments**
 - Posted online via **Canvas**
 - You will turn them into **Gradescope** by the due date (**USE THE TEMPLATE GIVEN TO YOU!!!**)
- Process:
 - I give you a PDF with the questions on it
 - You **edit** the PDF and insert your answers
 - Sometimes, you will write programs separately
 - You **save** your work
 - You **upload** the PDF (and, if applicable, program) onto **Gradescope**
 - **AGAIN: MAKE SURE YOU ONLY USE THE TEMPLATE GIVEN TO YOU** (or risk having your work ungraded)

Getting Help from Us

Do's of Office Hours:

- **Do** go to office hours regularly starting in the first week!
- **Do** prepare some specific questions: these may be related to the material or syllabus (*do* read the syllabus!)
- **Do** ask how the professor recommends approaching or preparing for the work you have to do
- **Do** talk about your aspirations or goals!
- **Do** feel comfortable asking the professor about how they got into their field and their professional life journey

Don'ts of Office Hours:

- **Don't** be too intimidated: you *really* can ask anything you like about the course (or CS, or industry, or grad-school, etc.)
- **Don't** expect to be given *direct* answers to the assignments

Getting Help from Us

- **Office Hours!**

- Are listed in syllabus and on Canvas
- Please come talk to us (we get lonely!!! )

- **Piazza!**

- We will check-in daily to help there
- Students should help each other out on Piazza too!!

- **Email!**

- *But only if you have to*, e.g. if it's personal.
- If you are emailing, **PLEASE** say it's ***CS64-related*** in the subject field

THIS WEEK!!!!

- There is a lecture on Wednesday 😐
- There IS an assignment for you on Canvas: **assignment01** due Mon 4/6 by 11:59pm
- There IS a reading for you on Canvas: **handout01**
 - *Please read the entire thing!*
- There IS a lab section this week on **Friday!**

I will now answer any questions you have
re: the course! 😊

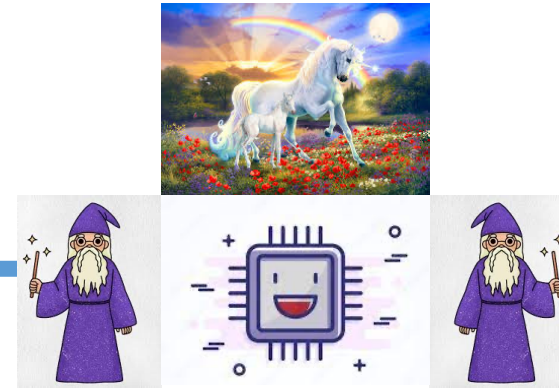
...and then we'll move on to the 1st lesson!
Basic Computer Concepts and Numerical Bases

Computer Organization aka The “Digital Mechanisms” in the CPU

- This is an **introductory** course in **low-level programming** and in **computer hardware design**.
 - Two separate but very intertwined areas
- What happens between your C/C++ command:
int a = 5, b = 7, c = a + b;
 - and the actual “**digital mechanisms**” in the CPU that processes these “simple” (and other “no-so-simple”) commands?



So Why Digital Design?



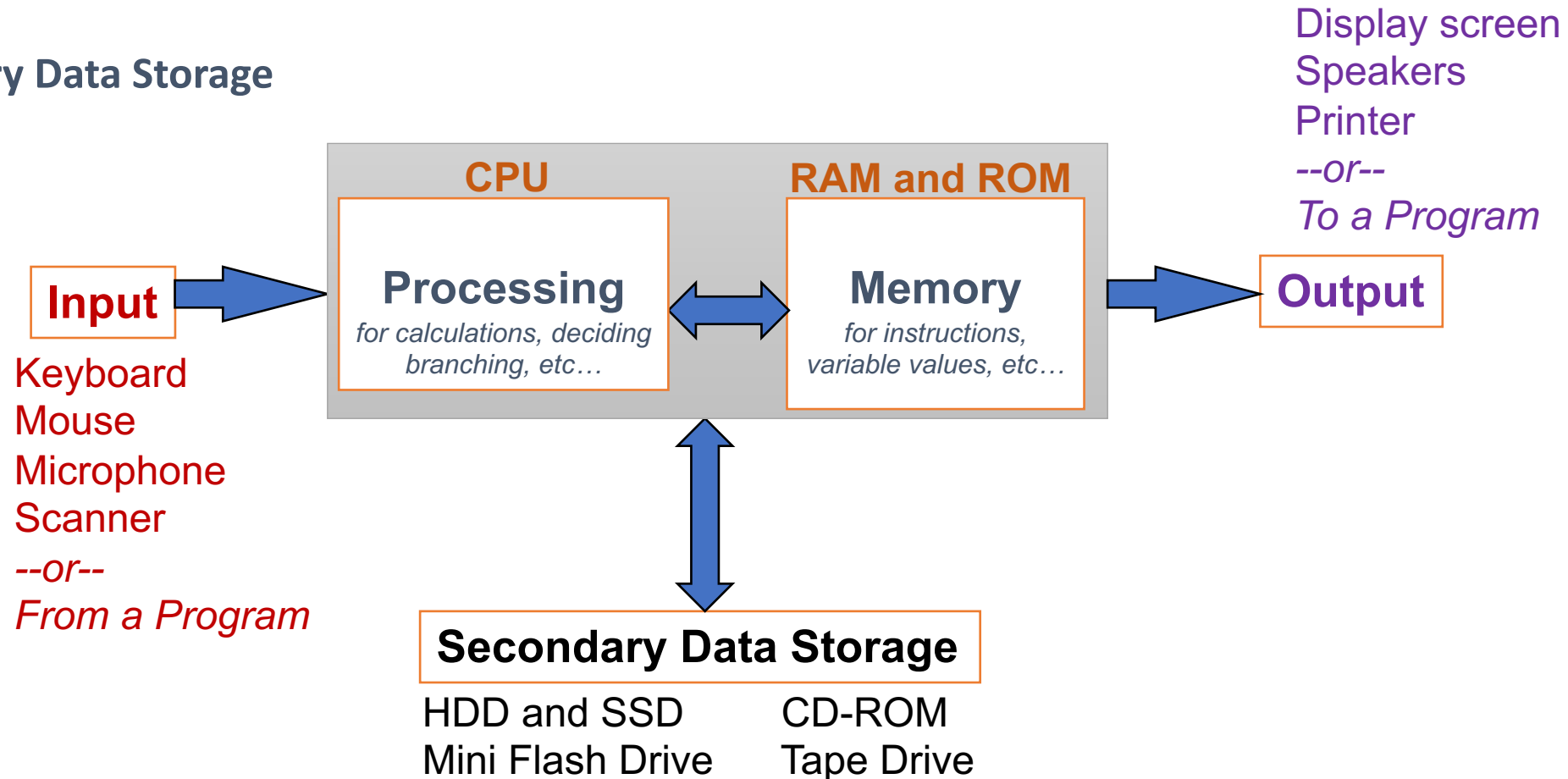
- Because that's where the “magic” happens
- Logical decisions are made with 1s and 0s and are run by the hardware
- Physically, this comes from
 - How electrical currents “switch” one way or the other inside semiconducting material
 - *But we don't have to worry about the physics in this class...! (leave that to EE/ECE)*
- We will *not* go into “*deep*” dives with digital logic designs in this course
 - We will, however, learn about the ***fundamental*** workings of digital logic designs and how they are used for ***computing purposes***.

A Simplified View of Modern Computer Architecture

The 5 Main Components of a Computer:

1. Processor
2. Memory
3. Input
4. Output
5. Secondary Data Storage

a.k.a von Neumann Architecture



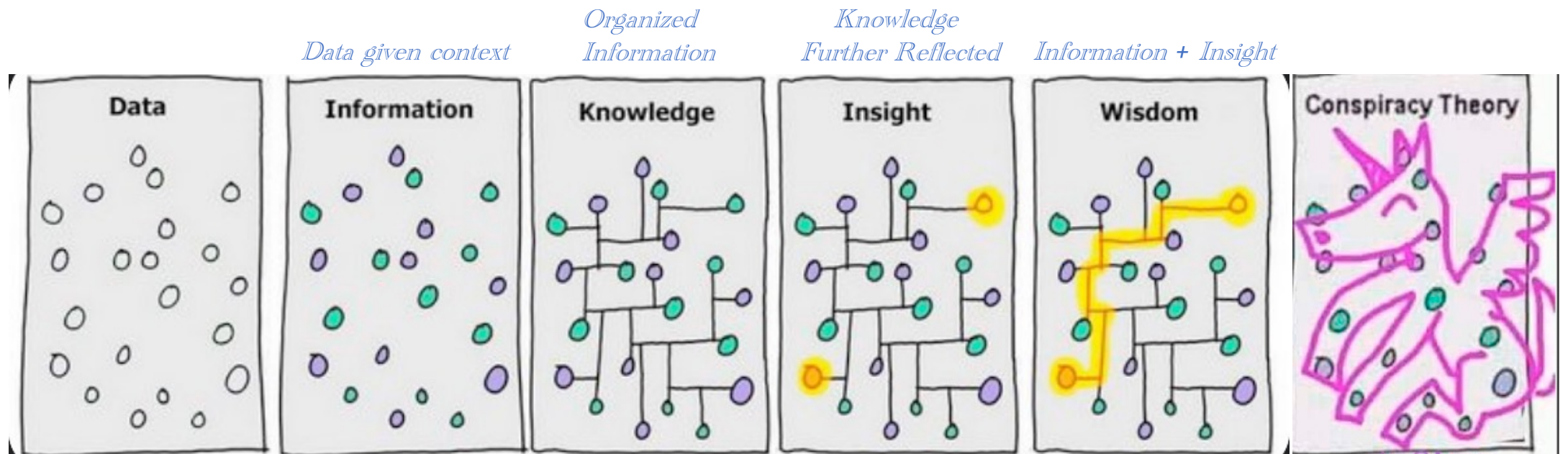


Image from: Facebook page for **Scientists for EU**, 11/4/20
<https://www.facebook.com/scientistsforeu/posts/3030750643693591>

Terms We Should be *Comfortable* With

- **Binary Values**

- Base-2 integers, i.e. numbers made up of “0” and “1” digits...
- What ALL data and ALL instructions are made of in a computer!

- **Data**

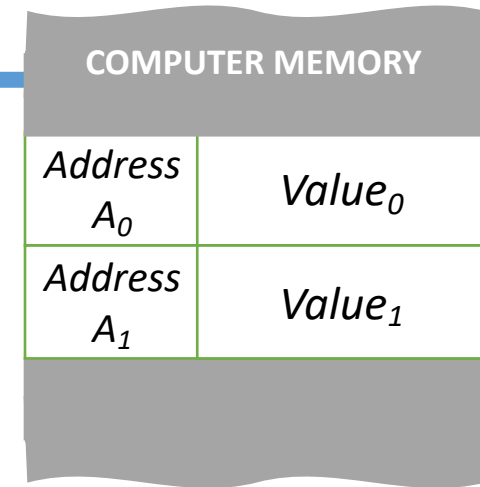
- Useful values in a computer:
- Values of variables (e.g. in a program) ; of locations of values (i.e. *addresses* in computer *memory*)

- **Instruction**

- A message (information) that tells the computer *processor* what to do with the data
- Many instructions = a *program* (BTW, are *algorithms* the same thing?)

Computer Memory

- Memory is usually organized in 2 parts:
 - Data Address: **Where** can I find my data?
 - Data Value: **What** is my data?
- The smallest representation of the data
 - A binary *bit* (“0”s and “1”s) **8 bits = 1 byte**
 - A common collection of bits is a *byte*
- **What is the *minimum* number of bits needed to convey an alphanumeric character? And WHY?**



COMPUTERS ARE DIGITAL MACHINES

THEY ARE DESIGNED
TO COUNT IN...

2s

Counting Numbers in Different Bases

- We “normally” count in 10s
 - Base 10: **decimal** numbers
 - We use *10* numerical symbols in Base 10: “0” thru “9”
- Computers count in 2s
 - Base 2: **binary** numbers
 - We use 2 numerical symbols in Base 2: “0” and “1”
 - Represented with 1 bit (Note: $2^1 = 2$)

Counting Numbers in Different Bases

Other convenient bases in computer architecture are “versions” of binary:

- Base 8: **octal** numbers
 - Number symbols are 0 thru 7
 - Represented with **3 bits** (note how $2^3 = 8$)
- Base 16: **hexadecimal** numbers (aka **hex**)
 - Number symbols are 0 thru F:
including **A = 10, B = 11, C = 12, D = 13, E = 14, F = 15**
 - Represented with **4 bits** (note how $2^4 = 16$)
 - **Why are 4 bit representations convenient???**

Positional Notation

642 in base 10 (**decimal**) can be described in “**positional notation**” as:

$$\begin{aligned} 6 \times 10^2 &= & 6 \times 100 &= 600 \\ + 4 \times 10^1 &= & 4 \times 10 &= 40 \\ + 2 \times 10^0 &= & 2 \times 1 &= 2 \end{aligned} \quad = 642 \text{ in base 10}$$

6	4	2
10^2	10^1	1

$$642_{\text{(base 10)}} = 600 + 40 + 2$$

Positional Notation

$$\sum d_n B^n$$

This is how you convert **any** base number into decimal!

Each digit (d_n) gets multiplied by B^n

Where:

B = the base

n = the position of the digit

1

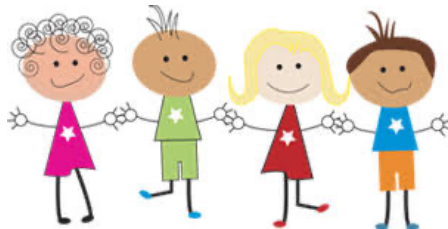
position
0

Example: given the number **642** in **base 8**:

Number in decimal = $6 \times 8^2 + 4 \times 8^1 + 2 \times 8^0 =$ **418** in decimal

It's Convenient to Remember the Powers of 2

Power of 2	Value
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1024



Another Convenient Table...

HEXADECIMAL	BINARY
0x0	0000
0x1	0001
0x2	0010
0x3	0011
0x4	0100
0x5	0101
0x6	0110
0x7	0111
0x8	1000
0x9	1001

HEXADECIMAL (Decimal)	BINARY
0xA (10)	1010
0xB (11)	1011
0xC (12)	1100
0xD (13)	1101
0xE (14)	1110
0xF (15)	1111

Note: the convention for writing hex numbers is to put **0x** before the number.

YOUR TO-DOs

ADDING THE COURSE?
TALK TO ME NOW!

* Remember to Check Piazza every day! 😊

- Before Wednesday's Lecture:

1. Check that you have accounts on **Canvas**, **Piazza**, **Gradescope**, and **CoE for Unix**
 2. **Get iClicker account!**
 3. Read the syllabus! 😊
 4. **Do your reading**
(Handout #1 – at least Sections 1, 2, and 3 for Wednesday)
- Look for Assignment #1 on Canvas
 - Will be due next Monday, will also discuss this week in lab section

</LECTURE>